

Position 3383: Scientist (f/m/x) for Environmental Data and Modelling Systems

Technical homework exercise

EXPECTED EFFORT

Approximately 120 minutes in total

CONTACT PERSON

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SUBMISSION

Please bring your solution to the interview

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Please complete both tasks. Each task is intentionally limited in scope and should take approximately 60 minutes. We value clear reasoning and sensible prioritization over completeness. If you stop before addressing every point, briefly describe what you would do next.

Use of AI tools

You are welcome to make extensive use of AI and your usual toolkit. Please briefly state which tools you used and how they contributed to your solution. We are interested in your product and engineering judgment, your assumptions, and your verification process.

TASK 1

Backend maintenance: validating route geometry

Expected effort: approximately 60 minutes

Context

A route is represented as an ordered list of GPS points. A valid line geometry must contain at least two points. The backend currently attempts to create a line when only one point is provided.

The relevant shared service is [GeoApiService.php](#).

Your task

Update the implementation so that line creation rejects invalid input with a clear error message. Add or adjust automated tests.

Please also provide a short note describing:

- what you changed and why;
- how you verified the change;
- additional edge cases you noticed and whether you handled them;
- any assumptions or tradeoffs you made.

You are not expected to understand the entire codebase. Keep the change focused. A partial but well-reasoned solution is acceptable within the time box.

TASK 2

Refining transect creation on mobile

Expected effort: approximately 60 minutes

Context

In a biodiversity-monitoring mobile app, users enter field observation data while working outdoors.

A **transect** is a sampling route consisting of multiple ordered geographic points. It belongs to a monitoring site. Ecologists walk along transects while recording occurrences of individual birds, beetles, or other species.

Initial requirement

A logged-in user is viewing a map screen. Existing transects may be visible on the map. The user should be able to create a new transect from this screen.

Your task

Refine this initial product requirement so that it is ready for implementation and QA. Do not write code or build a working prototype. The requirement intentionally leaves questions open: show us how you identify and handle them.

Please describe:

- the proposed user journey for creating a transect;
- questions you would ask product owners, domain experts, or engineers;
- assumptions you make where answers are unavailable;
- acceptance criteria that an engineer and QA tester could use;
- relevant validation rules, edge cases, and error states;
- tradeoffs or alternative approaches you considered.

Format

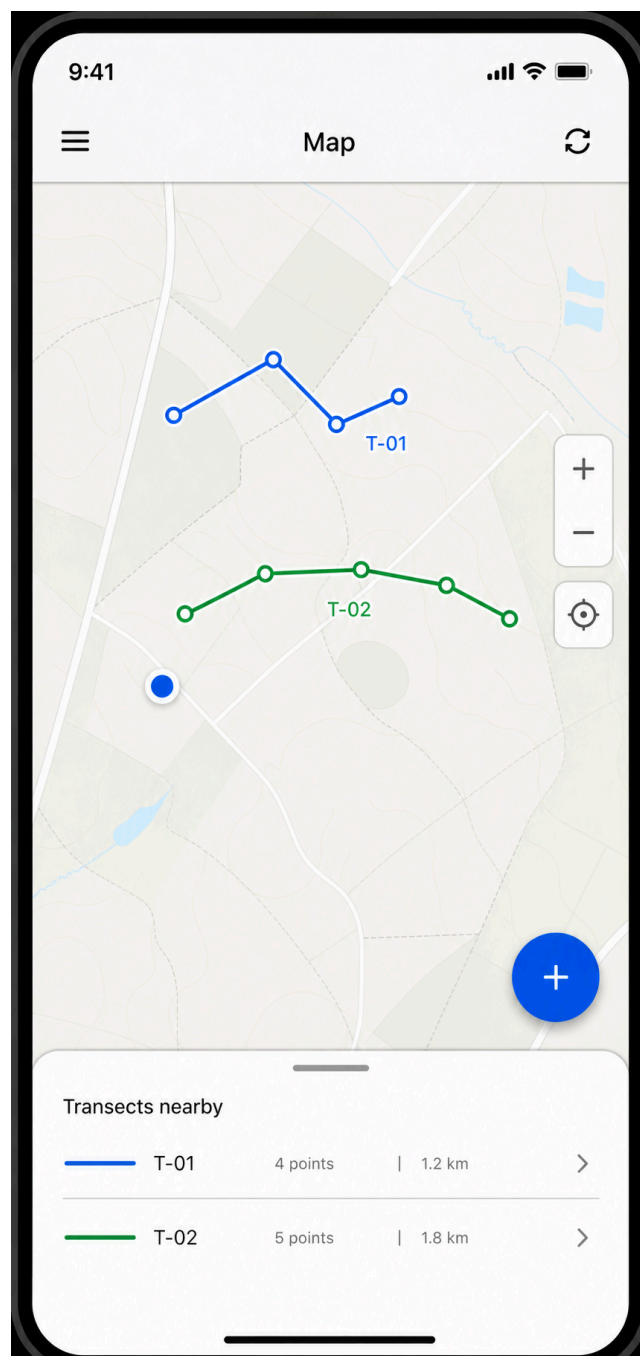
Use any format that helps communicate your proposal: annotated sketches, Figma screens, AI-generated wireframes, written UX descriptions, or a combination of these. Visual polish is not part of the assessment.

The next page contains an illustrative map screen as a starting point. It is not intended to answer open questions or constrain your proposal.

TASK 2: ILLUSTRATIVE WIREFRAME

Starting point: map screen

The following screen is an illustrative starting point for the exercise. It is not intended to answer open questions or constrain your proposal.



Existing transects T-01 and T-02 are visible as polylines with ordered geographic points. Consider this a wireframe rather than a prescribed final design.